CRYONANO

Variable Temperature Insert (VTI)

High-Field Air-Core Electromagnet with Advanced Thermal Management

Controlled Cryogenic Operation.
Stable Vacuum Performance.
Built for Research Applications.

The CRYONANO Cryogenic Dipstick & LN_2 Insert is a modular cryogenic platform designed for precise low-temperature control and electrical measurement in advanced research and industrial environments.
Engineered for operation from liquid nitrogen temperatures (~77 K) to above ambient, the system integrates OFHC copper cold stages, vacuum-compatible electrical feedthroughs, and closed-loop temperature control, enabling stable and repeatable measurements in cryostats, vacuum chambers, and custom experimental setup.

KEY FEATURES

- **Thermal Performance**
  Operates from 77 K to 325 K with rapid thermal equilibration via OFHC copper cold stage, minimized sample temperature gradients, and stable long-duration performance.
- **Electrical & Signal Handling**
  24-pin vacuum feedthrough supports high-voltage lines, sensor signals, thermocouples, and measurement signals while maintaining signal integrity in cryogenic and vacuum environments.
- **Mechanical & Vacuum Design**
  High-vacuum compatible construction with integrated radiation shielding, precision-polished mounting surfaces, and replaceable fixtures for fast sample exchange.
- **Temperature Control**
  Integrated PT1000 sensor, embedded ceramic chip heater, and closed-loop PID regulation support fixed, stepped, and variable temperature modes.

Notes

- Designed for vitrified sample preservation and cryogenic workflows
- Requires proper LN_2 handling and cryogenic safety procedures
- Performance dependent on microscope compatibility and setup

APPLICATIONS

- **Low-temperature electrical measurements**
- **Material characterization at cryogenic temperatures**
- **Semiconductor device testing**
- **Sensor calibration and evaluation**
- **Cryogenic physics experiments**
- **High-voltage measurements in vacuum**

Variable Temperature Insert (VTI)

SPECIFICATIONS

THERMAL & CRYOGENIC PERFORMANCE	
Operating Temperature Range	77 K to 325 K (expandable ~70–358 K)
Cooling Method	LN_2 dip / optional flow channel / cold finger
Cold Stage Material	OFHC Copper
Thermal Conductivity	> 390 W/m·K
Radiation Shielding	Integrated

SAMPLE & MECHANICAL	
Sample Size Compatibility	Up to 20 mm × 20 mm (custom supported)
Surface Finish	< 0.1 µm Ra (precision polished)
Mounting	Replaceable fixtures / clamp-based
Flange Options	ISO-KF / CF / custom

TEMPERATURE CONTROL SYSTEM	
Sensor Type	PT100 / PT1000 RTD
Sensor Placement	Near sample mount
Heater Type	Embedded ceramic chip heater
Heater Supply	24 V DC ±5%, up to ~2.2 A
Control Method	External PID controller

ELECTRICAL & FEEDTHROUGH	
Feedthrough Type	24-pin vacuum feedthrough
Voltage Rating	Up to 500 V DC per pin
Current Rating	~1 A per pin
Signal Types	Power, sensors, thermocouples, test signals
Insulation Resistance	> 10^9 Ω @ 500 VDC
Wiring	Thermally anchored twisted pairs

VACUUM COMPATIBILITY	
Sensor Type	PT100 / PT1000 RTD
Sensor Placement	Near sample mount
Heater Type	Embedded ceramic chip heater
Heater Supply	24 V DC ±5%, up to ~2.2 A
Control Method	External PID controller

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